arXiv:2611.09794v1 [cs.RO] 6 Nov 2026

# Seeing What Matters: Benchmarking Task-Conditioned Actionable Spatial Belief for Manipulation

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## Abstract

*When should a robot look at the cup, and when at the portafilter? Spatial-reasoning benchmarks evaluate whether a model can describe a scene, but manipulation needs something more specific: the ability to acquire the particular evidence that makes the current command executable. A view that is informative for one task can be useless for another, even at the same workstation. We introduce Seeing What Matters, a benchmark for task-conditioned actionable spatial belief in simulated tabletop manipulation. Across three domains—coffee workstation, drawer/cabinet, and microwave/appliance—the benchmark provides 860 frozen evaluation cases; expanding across methods, budgets, and inference repeats yields more than 115,000 evaluator episodes under two protocols: closed-loop actionability and same-object functional out-of-distribution generalization. Each case pairs a set of candidate RGB views with a manipulation command, shared scripted skills, and hidden postconditions, so the benchmark directly measures whether a view-selection policy acquires the observations its command actually needs. We also provide a diagnostic reference pipeline that maintains a spatial–functional belief state and ranks views by missing evidence. At primary budget settings, this pipeline reaches 98.8% closed-loop and 98.9% functional-OOD success; ablating the functional or command-binding components collapses performance to near zero. Seven vision-language models, evaluated on the same cases, reach only 26–43%. Replacing the reference pipeline's coverage metadata with frozen RGB-derived scores preserves 98.5%/98.9% success; shuffling the visual scores collapses it to 15%/11%, confirming that the result depends on image content. These gaps suggest that spatial belief for manipulation should be evaluated not by scene-level completeness, but by whether it acquires the evidence the current command requires.*

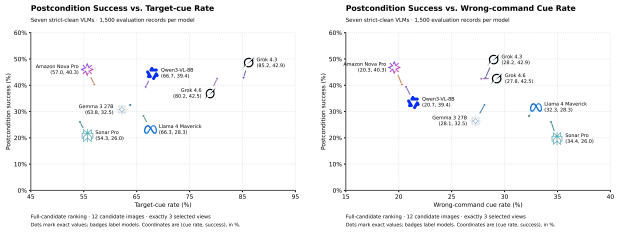



Figure 1. **Seven-model evidence diagnostics.** Left: success versus target-cue rate. Right: success versus wrong-command cue rate. Seven strict-clean VLMs each have 1,500 evaluation records, preview twelve RGB candidates, and select exactly three views. Dots show point estimates; icons label models. Cue rates are separate evaluator diagnostics, not complementary fractions.

## 1. Introduction

An embodied agent can look at the right scene and still acquire the wrong evidence. Consider a coffee workstation: tamping requires observing the portafilter basket, puck surface, and tamper alignment; cup placement requires cup–spout alignment, support, and clearance; grinding requires the hopper, chute, receiver, and relevant controls. The physical workspace may be unchanged, yet the observation that makes one command executable can be irrelevant to another. Seeing more is not necessarily seeing what matters.

Recent work has expanded spatial reasoning from passive question answering toward active belief construction and embodied interaction [10, 39]. Theory of Space studies whether agents can acquire, maintain, and revise spatial beliefs through exploration [49]; ESI-Bench evaluates spatial intelligence through perception–action loops [12]; and BEHAVIOR-Spatial connects spatial understanding and instruction following to downstream robot behavior [6]. In parallel, vision-language-action policies and affordance-centered methods increasingly ground language in object

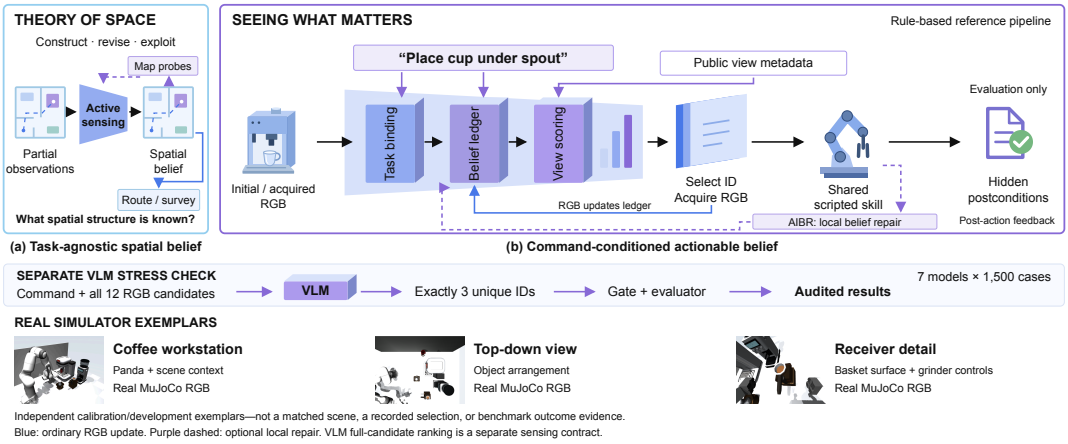



Figure 2. **From spatial belief to actionable evidence.** Theory-of-Space-style evaluation asks whether an agent can construct and revise spatial beliefs through exploration. Seeing What Matters keeps the active-belief motivation but scores whether command-conditioned views expose the evidence needed by a shared scripted skill, with hidden postconditions used only for evaluation. Schematic modules are illustrative.

parts, action regions, and manipulation trajectories [4, 19, 25, 29, 48], while active-vision systems learn where to look before acting [1, 2, 18, 21, 32, 41, 44].

These directions motivate a narrower but consequential question: *given a command, can an agent acquire the particular spatial-functional evidence required to execute it?* Existing evaluations may measure the accuracy of a spatial answer, the quality of a global representation, or the final success of a learned policy. Final success alone, however, does not reveal whether an agent selected the correct evidence, relied on a shortcut, or was rescued by the controller. Conversely, a geometrically accurate belief may contain many facts while omitting the one relation that makes the next action safe and executable. A controlled evaluation should therefore hold the scene, candidate observations, and downstream skill fixed, vary the command and observation budget, and test whether the acquired evidence supports a hidden task outcome.

We introduce *Seeing What Matters*, a frozen RGB benchmark for *task-conditioned actionable spatial belief* (Fig. 2). The benchmark spans three simulated tabletop domains: a five-task coffee workstation (500 cases covering tamping, portafilter insertion, cup placement, serving, and grinding), a drawer/cabinet interface (180 cases emphasizing handle access, openness, and reachability), and a microwave/appliance interface (180 cases emphasizing cavity support and control state)—860 evaluation cases in total (Figs. 3 and 4). Each case pairs a command with twelve frozen candidate-view RGB observations, a shared scripted skill interface, and hidden simulator-state postconditions; the resulting base contains 6,000 candidate-view records. Expansion across evaluated methods, observation budgets, and repeated runs produces more than 115,000 evaluator episodes, including 60,000 closed-loop and 55,500 same-object functional-OOD episodes from the primary coffee domain. The functional-OOD episodes preserve object identity while changing command-conditioned functional requirements, testing whether a selector follows the requested task rather than an object- or scene-level shortcut.

The evaluation separates two sensing contracts. Under the primary sequential protocol, every policy starts from the same frozen candidate set and initial state, acquires observations under a common budget $H$, and passes the resulting evidence to the same scripted skill. Under a separate strict-clean VLM stress test, a model previews all twelve frozen RGB candidates and must return exactly three ranked view identifiers; this second experiment evaluates full-candidate visual ranking rather than an equivalent active-acquisition policy. Both protocols reserve simulator-state postconditions for evaluation: hidden state is never supplied to the view selector. This separation makes it possible to attribute failures to evidence acquisition without conflating them with learned low-level control.

We additionally provide a transparent, metadata-assisted reference pipeline. A Spatial–Functional Belief Ledger records task-relevant pose, relation, function, provenance, confidence, and staleness fields. A rule-based View Belief Map binds the current command to required predicates and ranks candidate views by their expected gain over missing evidence; it uses the command, public candidate-view metadata, and the current ledger state, but not hidden postconditions or labels from unselected observations. Optional Action-Induced Belief Revision compares predicted and observed postconditions after execution, localizes potentially stale ledger entries, and requests targeted re-observation. The purpose of this pipeline is not to establish a general learned vision policy but to provide an auditable reference whose command-to-evidence decisions can be inspected di-

rectly.

The results expose a substantial gap between task-conditioned evidence selection and generic visual ranking. At the primary settings, the diagnostic reference pipeline reaches $98.8\%$ closed-loop success at $H = 5$ and $98.9\%$ functional-OOD success at $H = 3$. Removing functional belief fields reduces functional-OOD success to $0.0\%$; removing command binding reduces it to $19.7\%$. Under the more restrictive $H = 1$ initial-selection setting, closed-loop and functional-OOD success fall to $19.0\%$ and $58.6\%$, respectively. In the separate full-candidate stress test, seven strict-clean VLMs achieve only $26.0\%$–$42.9\%$ postcondition success despite previewing every candidate image (Fig. 1): models frequently detect target-related cues without consistently selecting evidence that supports the hidden task outcome. These findings suggest that actionable belief requires more than scene recognition: it requires binding the current command to the evidence needed for execution.

**Contributions.** Our contributions are threefold:

- We formulate *task-conditioned actionable spatial-functional belief*, evaluating a spatial belief by whether it exposes the parts, relations, and functional states required by the current command. We introduce a frozen RGB benchmark spanning three simulated tabletop domains (five-task coffee workstation, drawer/cabinet, and microwave/appliance) with 860 evaluation cases, 6,000 candidate-view records, shared scripted skills, hidden simulator-state postconditions, and controlled closed-loop and functional-OOD evaluation.
- We provide an auditable metadata-assisted reference pipeline comprising a Spatial–Functional Belief Ledger, a rule-based View Belief Map, and optional Action-Induced Belief Revision, while explicitly separating public selection information from hidden evaluator state.
- We evaluate budget-matched baselines, ablations, no-image and shortcut controls, paired scene-group bootstrap estimates, observation-cost behavior, and seven strict-clean VLMs, accompanied by manifests, raw records, table-generation artifacts, and traceable admission checks.

Figure 2 summarizes the central distinction: spatial-belief evaluation asks what an agent knows about a scene, whereas *Seeing What Matters* asks whether that belief helps the agent acquire the evidence made actionable by the current command.

## 2. Related Work

**Spatial intelligence and embodied evaluation.** Spatial-reasoning evaluation has advanced across complementary layers. Benchmarks now test spatial memory and measurement from video, egocentric embodied cognition, and reference frames and spatial relations in 2D/3D observations [9, 35, 46]; diagnostic and comprehensive evaluation suites probe spatial-reasoning breadth [40, 42]; and vision-language methods train spatial reasoning, optimize 3D layouts, and support agentic spatial understanding [5, 23, 36]. EmbodiedQA and Habitat established goal-directed navigation and interactive evaluation in simulated environments, while ENACT evaluates egocentric interaction through world modeling [10, 30, 39]. XVR trains cross-view correspondence, verification, and localization and transfers the resulting representations to manipulation [15]. Theory of Space asks whether an exploring agent forms and revises an internal spatial belief, connecting modern foundation-model evaluation to classical planning under partial observability [17, 49]. These works principally score answers, representations, or broad agent behavior. We focus on a narrower controlled interface: whether observations selected *before execution* expose the command-specific evidence required for a shared manipulation skill to satisfy a hidden postcondition.

**Manipulation benchmarks and policies.** RLBench, ManiSkill2, and CALVIN provide standardized task suites for learning and evaluating diverse or long-horizon manipulation policies [11, 14, 24]; CLIPort and VIMA further test language- or multimodal-prompt-conditioned tabletop generalization [16, 34]. ALFRED and TEACh evaluate language-grounded household task execution, with TEACh additionally requiring interactive dialogue [26, 33]. SAPIEN, Habitat, and BEHAVIOR-1K provide complementary simulation infrastructure and activity suites for controlled embodied evaluation [20, 30, 43]. These settings primarily vary the learned policy and score end-to-end task completion or generalization. Our benchmark instead fixes the downstream skill and hidden evaluator, isolating whether a method first acquires the command-specific spatial-functional evidence that makes execution possible; this decomposition complements, rather than replaces, evaluation on broad manipulation suites. Within the learned-policy literature, Where2Act localizes actionable regions on articulated objects, SeqAfford grounds sequential 3D affordances, and RT-2, OpenVLA, SpatialVLA, and $\pi_0$ map visual-language context to robot actions or spatially grounded action representations [3, 4, 19, 25, 29, 48]. ActiveVLA and SaPaVe explicitly incorporate critical-region localization or active view–pose reasoning [21, 22]. PhysVLM focuses on physical reachability, while RoboGround grounds manipulation with vision-language priors [13, 51]. GraspGPT and ThinkGrasp use language-model or vision-language reasoning for task-oriented grasping and strategic part grasping in clutter [28, 37].

**Active perception for embodied action.** Classic active-vision and next-best-view work selects sensing actions to reduce uncertainty or improve reconstruction [1, 2, 8, 27, 31]. Recent robotic systems learn task-directed gaze, camera mo-

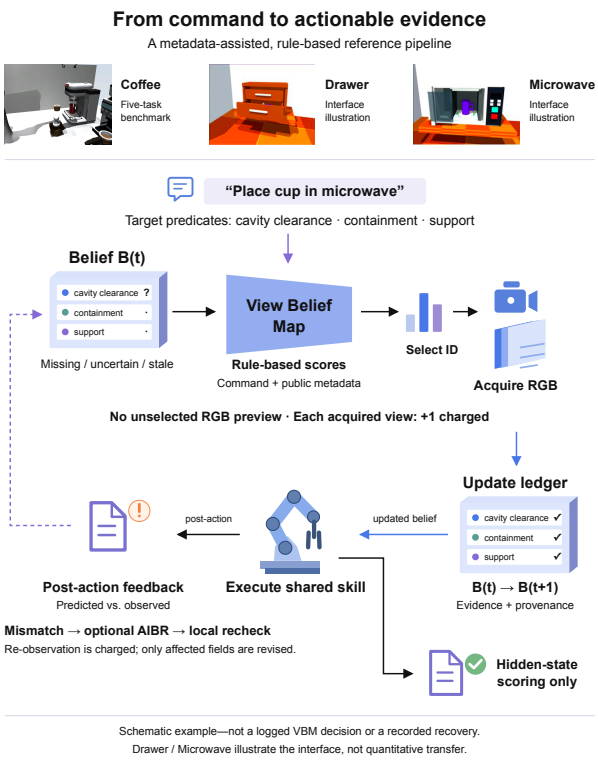



Figure 3. **Task-conditioned evidence acquisition.** The coffee workstation supplies the quantitative five-task benchmark; Drawer/Cabinet and Microwave/Appliance are interface illustrations whose small quantitative sanity checks appear in the supplement. (a) The command and ledger identify missing evidence. (b) The transparent View Belief Map ranks candidate views. (c) Acquired observations update the ledger before a shared scripted skill executes. (d) Post-action observations support optional local revision; hidden simulator state is reserved for evaluation. This hybrid schematic combines abstract icons with real MuJoCo RGB exemplars; the exemplars are illustrative interface views, not recorded selections or outcome evidence.

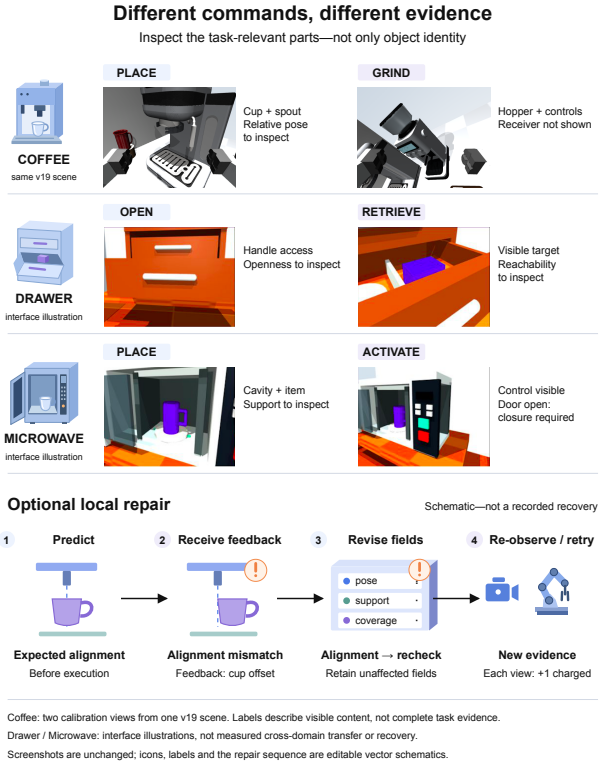



Figure 4. **Commands determine useful evidence; mismatches motivate local revision.** (a) Coffee placement requires cup–spout alignment, whereas grinding requires the hopper–chute–basket chain. (b) Drawer opening and retrieval emphasize handle access, openness, target visibility, and reachability. (c) Microwave placement and activation emphasize cavity support and door/control state. (d) An illustrative coffee-placement mismatch triggers rechecking of alignment while retaining support evidence, followed by re-observation and retry. This hybrid schematic combines abstract annotations with real MuJoCo RGB exemplars; highlighted regions are not measured detections, and the repair example does not establish cross-domain recovery performance.

tion, or base placement: Vision in Action learns human-like search and focus from demonstrations; EyeRobot couples gaze with hand actions; Observe Then Act coordinates camera next-best-view selection with gripper next-best-pose planning; and Mobi-$\pi$ chooses robot base poses that make a downstream policy effective [18, 38, 44, 45]. VISO-Grasp and Observer-Actor further couple active view planning or sparse-view perception to manipulation [32, 41]. WoMAP likewise uses learned world models for active open-vocabulary localization under partial observability [47]. We share their task-directed sensing motivation, but target a different evaluation question: whether a fixed candidate-view selector, under matched budgets and with the same downstream skill and evaluator for every method, acquires the right evidence.

**Closest concurrent benchmarks.** FUSE studies learned semantic–geometric evidence acquisition for functional affordance grounding, ESI-Bench broadens embodied spatial evaluation to perception–action loops, VABench studies visual demonstrations, active perception, and metric control, and BEHAVIOR-Spatial connects spatial understanding and instruction following to robot behavior [6, 7, 12, 50]. These concurrent efforts broaden the learned-policy or agent setting. Seeing What Matters contributes a controlled complement: frozen RGB candidates, explicit sensing contracts, command-shift interventions, a fixed skill interface, hidden simulator-state postconditions, and a metadata-assisted reference selector whose decision trace can be audited. This narrower design is intended to isolate *which evidence was acquired and why*, not to establish a general-purpose active-

vision policy.

## 3. Benchmark and Method

**Task interface.** Each episode contains a hidden simulator state $s$, a manipulation command $\tau$, a set of candidate views $\mathcal{V}$, and RGB observations acquired under budget $H$. The agent sees the command, acquired RGB frames, candidate-view metadata, and its own ledger state; it never sees hidden predicates, unselected endpoint renders, evaluator labels, contact logs, or postcondition ground truth before acting. The five task families—tamp, insert, place, serve, and grind—each have 50 scene groups and two variants, yielding 500 frozen cases and 6,000 rendered candidate-view RGB records. All downstream execution uses the same shared scripted skill library, so differences in task outcome trace to evidence acquisition rather than controller quality.

**Evaluation protocols.** The benchmark defines two protocols that isolate different aspects of evidence acquisition. Protocol C.1 (*closed-loop actionability*, primary budget $H{=}5$) tests sequential view selection: the agent acquires up to five views one at a time, updating its ledger after each observation, then passes the accumulated evidence to the scripted skill. Protocol C.2 (*same-object functional OOD*, $H{=}3$) holds object identity fixed across commands and varies only the functional requirements—a cup that must be *placed* needs alignment and clearance evidence, whereas the same cup in a *serve* command needs fill-level and stability evidence. This split exposes whether a selector follows the commanded task or falls back on an object- or scene-level shortcut. In both protocols, hidden simulator-state postconditions are evaluated only after execution: the evaluator checks whether the physical outcome satisfies the command's requirements without revealing those conditions to the agent beforehand.

**Spatial–functional belief ledger.** The Spatial–Functional Belief Ledger stores entries

$$B_t = \{(p_i, \hat{y}_i, q_i, E_i, \kappa_i, \rho_i)\}_{i=1}^{N_t},$$

where $p_i$ is a spatial or functional predicate, $\hat{y}_i$ is its estimated value, $q_i$ is confidence, $E_i$ records visual provenance, $\kappa_i$ marks staleness, and $\rho_i$ marks task relevance. Predicates include part visibility, relative pose, support, containment, action affordance, functional cue, and predicted postcondition state. In the reference implementation, $\rho_i$ is instantiated from command templates, $E_i$ stores the selected view IDs that support an entry, $q_i$ is a three-level confidence flag (observed, inferred, or unknown), and $\kappa_i$ is raised only by actions that may invalidate the corresponding predicate type. The benchmark evaluates whether a policy acquires the command-relevant evidence it needs; the ledger's role is to make that evaluation transparent and traceable, not to serve as an optimal learned state estimator.

**View Belief Map.** For each candidate view $v$, the agent estimates which missing target predicates the view may resolve:

$$J(v; B_t, \tau) = \frac{1}{|\mathcal{M}_t|} \sum_{p_i \in \mathcal{M}_t} g(v, p_i, \tau) - \lambda\, c(v),$$

where $\mathcal{M}_t$ is the set of missing, uncertain, or stale target-relevant predicates, $g$ is an ordinal evidence-gain estimate, and $c(v)$ is normalized view cost. The selected views form a discrete View Belief Map over the candidate camera set—a command-conditioned score over executable observations, not a dense point-cloud value field.

The key implementation detail is that $g$ is a deterministic ordinal score, not a learned RGB scorer or privileged predicate checker. It uses the command template $R(\tau)$, the ledger's missing/stale set $\mathcal{M}_t$, and public candidate metadata including semantic view type, camera relation, and nominal functional-part coverage. This metadata is a strong design aid and is disclosed as part of the reference method; it does not include hidden state, success labels, future postconditions, contact logs, or unselected RGB content. The resulting pipeline—hereafter the *diagnostic reference selector*—therefore serves as an auditable metadata-assisted anchor that future learned view selectors can be measured against. Figure 5 gives a concrete logged example of the score–select–execute–verify chain.

**Optional action-induced belief revision.** After a scripted skill executes, the evaluator returns a postcondition observation. Action-Induced Belief Revision compares predicted and observed postconditions, localizes the stale predicate type, lowers confidence in contradicted ledger entries, and selects a local repair view. Recovery is charged in the view/step budget and counted only when the final simulator postcondition passes. Because successful view selection makes mismatches rare in the primary protocol, Action-Induced Belief Revision is treated as a repair diagnostic and stress-test component rather than the main source of the benchmark's performance gains.

## 4. Experimental Setup

**Setup.** The experimental unit is 500 frozen cases (five tasks, 50 scene groups, two variants each), each paired with twelve candidate RGB views. We freeze task, postcondition, method, candidate-view, and failure-code manifests before evaluation. All non-privileged baselines share the same initial RGB view, candidate-view list, camera metadata, initial budget, scripted skill library, simulator reset distribution, and hidden postcondition evaluator. Hidden simulator predicates,

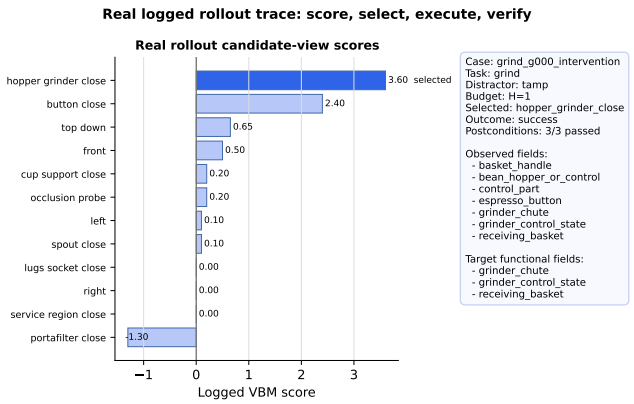



Figure 5. **Logged rollout trace for command-conditioned view selection.** This panel is generated from a real frozen functional-OOD trace. It shows the candidate-view scores, selected view, observed fields, target functional predicates, and hidden postcondition outcome for a successful one-view GRIND episode with a TAMP distractor command. The trace illustrates the audited decision that the benchmark scores: the policy succeeds by selecting the view that exposes the command-relevant functional chain before execution. This is one representative successful trace; hard $H{=}1$ aggregate success is reported in the experiments and supplement.

ground-truth postconditions, evaluator labels, and unselected endpoint renders are never exposed before action. The supplement specifies each method's sensing contract explicitly. Applying multiple methods, budgets, and inference repeats yields 60,000 closed-loop scripted-skill episodes and 55,500 functional-OOD episodes.

All raw records and SHA-256 manifests are retained.

**Statistical reporting.** At the primary budget, each task-method cell contains 300 episodes (50 scene groups $\times$ 2 variants $\times$ 3 inference repeats); each aggregate method row contains 1,500 episodes across five tasks. We treat frozen cases—not repeated answers from the same case—as the conceptual unit for generalization. The supplement reports paired scene-group bootstrap intervals over 500 case clusters, where each cluster is a task, group, and variant with three inference repeats averaged before resampling. Budget-curve bands reproduce source-provided episode-level normal-approximation 95% intervals and are descriptive. The appendix retains the full per-budget, per-task, bootstrap, and failure-diagnostic plots so that large aggregate gains can be checked against task-level and case-level behavior.

**Baselines.** Random samples valid views uniformly; Fixed scan follows a deterministic camera order; Pose coverage chooses geometry-revealing views regardless of command; History active reuses previously useful views without an explicit target ledger; Target ledger stores target predicates but does not score views by missing functional evidence; Privileged diagnostic uses privileged view labels under a fixed diagnostic selector. This privileged diagnostic is a privileged oracle-map/headroom anchor: it does not perform adaptive postcondition repair and may be affected by fixed single-policy tie-breaking. Success is the percentage of episodes whose scripted skill satisfies the hidden postcondition.

**Diagnostic QA versus actionability.** VLM question answering verifies whether selected images contain answerable spatial-functional information. It is not the main performance claim. The main evidence is whether selected observations make the downstream action pass simulator-state postconditions.

## 5. Results

**Closed-loop actionability.** Table 1 shows that task-conditioned view selection turns spatial belief into executable action. The diagnostic reference selector with optional Action-Induced Belief Revision reaches 98.8% average success, improving over History active by 45.9 points and over Target ledger by 86.4 points. The gain is not explained by memory alone: the ledger must expose which missing predicate should determine the next view. The budget curves (Fig. 9 in the supplement) confirm that the gap is not an artifact of choosing a single favorable budget; the advantage of the diagnostic reference selector persists across $H \in \{1, 3, 5, 8\}$ in both the closed-loop and functional-OOD protocols.

Shaded bands are episode-level normal-approximation intervals.

The near-ceiling primary results should not be read as the benchmark being solved under every observation regime. The supplement adds a hard $H{=}1$ budget stress split re-summarized from the frozen episode logs. At $H{=}1$, recovery-capable methods may still acquire additional charged observations after mismatch, so the constraint limits the initial selection, not the total observation cost. Even with that allowance, the diagnostic reference with Action-Induced Belief Revision drops to 19.0% in C.1 and 58.6% in C.2, exposing remaining difficulty that is hidden by the main $H{=}5$ and $H{=}3$ settings.

**Same-object functional OOD.** Table 1 evaluates the harder same-object setting. A method may observe the coffee machine and still fail if it observes the wrong functional part for the command. The diagnostic reference selector with optional repair reaches 98.9% success and 70.0 functional selectivity; category shortcuts and command-agnostic affordance policies fail despite seeing plausible object parts. The functional-evidence visualization (Fig. 13 in the supplement) shows the dissociation: target-functional cue observation

| Methods | *Task success (%)*↑ Tamp | Insert | Place | Serve | Grind | Avg.↑ | *Execution diagnostics* Coll.↓ | PostFail↓ | Views |
|---|---|---|---|---|---|---|---|---|---|
| **C.1: Closed-loop actionability** | | | | | | | | | |
| *Passive / generic policies* | | | | | | | | | |
| RANDOM | 3.7 | 15.3 | 14.7 | 17.0 | 4.7 | 11.1 | 6.2 | 82.7 | 5.0 |
| FIXED SCAN | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 8.1 | 91.9 | 5.0 |
| POSE COVERAGE | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 4.4 | 95.6 | 5.0 |
| HISTORY ACTIVE | 0.0 | 40.0 | 28.7 | **98.3** | 97.7 | 52.9 | 2.3 | 44.7 | 5.0 |
| *Belief policies* | | | | | | | | | |
| TARGET LEDGER | 18.3 | 0.0 | 43.7 | 0.0 | 0.0 | 12.4 | 3.1 | 84.5 | 5.0 |
| VIEW BELIEF MAP | **99.3** | 97.0 | 97.7 | 97.3 | 98.3 | 97.9 | 2.1 | **0.0** | 5.0 |
| VBM+AIBR | 98.7 | **99.3** | **98.0** | **98.3** | **99.7** | **98.8** | **1.2** | **0.0** | 5.0 |
| *Privileged diagnostic* | | | | | | | | | |
| PRIV. DIAGNOSTIC† | 98.7 | 99.3 | 97.3 | 97.3 | 97.7 | 98.1 | 1.9 | 0.0 | 5.0 |

| Methods | *Task success (%)*↑ Tamp | Insert | Place | Serve | Grind | Avg.↑ | *Functional evidence (%)* Target↑ | Wrong | Select.↑ |
|---|---|---|---|---|---|---|---|---|---|
| **C.2: Same-object functional OOD** | | | | | | | | | |
| *Category / command-agnostic policies* | | | | | | | | | |
| RANDOM | 4.0 | 3.7 | 9.3 | 4.0 | 2.7 | 4.7 | 32.9 | 32.6 | 0.3 |
| FIXED SCAN | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 20.0 | 20.0 | 0.0 |
| CATEGORY SHORTCUT | 0.0 | 0.0 | 18.3 | 0.0 | 0.0 | 3.7 | 26.7 | 26.7 | 0.0 |
| SPATIAL ONLY | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 16.7 | 23.3 | −6.7 |
| CMD-AGNOSTIC AFFORD. | 0.0 | 0.0 | 0.0 | 98.7 | 0.0 | 19.7 | 46.7 | 73.3 | −26.7 |
| *Command-bound belief policies* | | | | | | | | | |
| VIEW BELIEF MAP | **99.3** | **99.7** | **98.3** | 98.0 | **98.7** | 98.8 | **100.0** | 30.0 | **70.0** |
| VBM+AIBR | **99.3** | 99.0 | **98.3** | **99.0** | **98.7** | **98.9** | **100.0** | 30.0 | **70.0** |
| *Privileged diagnostic* | | | | | | | | | |
| PRIV. DIAGNOSTIC† | 98.3 | 98.3 | 98.7 | 99.3 | 99.0 | 98.7 | 100.0 | 30.0 | 70.0 |

Table 1. **Main results under both evaluation protocols.** *C.1* (closed-loop, $H{=}5$): baselines share the same candidate set and scripted skill interface. *C.2* (functional OOD, $H{=}3$): the same coffee-machine scene admits multiple commands; Target and Wrong report functional-cue hits; Select. is their difference. AIBR occasionally trades a small per-task decrement (e.g. Tamp in C.1) for cross-task repair capability. **Bold**: best; underline: second-best (non-privileged). †Fixed label-assisted calibration anchor, not an oracle upper bound.

tracks success, whereas wrong-command cue observation does not.

**Foundation-model stress test.** We upgrade the foundation-model view-scoring protocol from a 500-row endpoint stress check to a 1,500-case strict-clean benchmark. A VLM is admitted only when all 1,500 raw responses have HTTP 200/endpoint-OK status, strict JSON parses, unique exact-$H{=}3$ selections drawn from the twelve frozen candidates, complete evaluator records, exact case order, and no fallback parsing. Seven models pass this gate. Their postcondition success ranges from 26.0% to 42.9%: Grok 4.3 leads at $42.9 \pm 2.5\%$, followed by Grok 4.6 at $42.5 \pm 2.5\%$, Amazon Nova Pro at $40.3 \pm 2.5\%$, Qwen3-VL-8B at $39.4 \pm 2.5\%$, Gemma 3 27B at $32.5 \pm 2.4\%$, Llama 4 Maverick at $28.3 \pm 2.3\%$, and Sonar Pro at $26.0 \pm 2.2\%$ (normal-approximation 95% intervals; $n{=}1{,}500$ per model). All seven strict-clean VLMs cluster within a 16.9-point band across multiple provider and model families. Even the strongest model remains 55.9 points below the diagnostic reference selector at the primary setting, suggesting a systematic gap in command-conditioned view selection rather than an idiosyncratic weakness of one provider. The gap is not a controlled architecture comparison—the methods use different priors and sensing contracts—but it shows that current prompted ranking does not recover the benchmark's command-to-evidence rules.

The task-level pattern is consistent across models: Tamp and Place, whose target predicates involve large, visually salient parts (portafilter basket, cup–spout alignment), admit near-perfect scores from several VLMs, whereas Insert, Serve, and Grind require observing smaller or partially occluded functional chains (hopper–chute path, drip-tray clearance, grinder–receiver contact) that prompted VLMs fail to prioritize. This suggests that the gap is driven by the models' inability to bind the command to the specific functional-part evidence chain, not by a blanket failure to parse candidate images.

Only models passing the full 1,500-case strict-clean gate enter the reported table; the supplement documents the gate criteria and lists models whose endpoints were unavailable.

**Sanity checks.** The sanity-checks table (Tab. 5 in the supplement) consolidates the shortcut and leakage sanity checks. These controls are intentionally conservative: they reuse the same frozen same-object functional-OOD cases rather than introducing easier auxiliary tasks. Category shortcuts, command-agnostic affordance viewing, removing functional fields, and removing target conditioning all fail under the same hidden evaluator. This pattern argues against a leakage explanation in which success comes from object category, hidden postconditions, or a template rule that ignores the

command.

**Ablation.** The ablation table (Tab. 15 in the supplement) confirms that the decisive components are command binding, functional fields, and target conditioning. Removing functional fields does not harm the easier C.1 closed-loop protocol, but it collapses C.2 functional-OOD success to 0.0%; the distinction supports the claim that functional evidence matters specifically when object category is an unreliable shortcut. Removing Action-Induced Belief Revision has little effect in the primary functional-OOD split because mismatches are already rare after selecting the correct functional evidence; the recovery stress test (Fig. 14 in the supplement) isolates the setting where repair matters. The supplement reports the full task-level ablation matrices.

**RGB-derived coverage.** To test whether the reference pipeline depends on simulator-derived view metadata at inference, we replace its nominal view-to-predicate coverage mapping with VBM-VISION. A frozen CLIP-B/32 encoder first converts each candidate RGB image into a task-agnostic visual signature, and a lightweight per-predicate linear head estimates functional-part visibility. The head is trained with visibility labels on training scene groups and evaluated five-fold out of fold; every reported evaluation image is scored by a model that did not observe its scene group. At inference, the visual stage receives neither the command nor camera/view identifiers, simulator state, success labels, evaluator outputs, or nominal coverage metadata. The unchanged command-to-predicate binding is applied only after visual coverage prediction. With the evaluator, scripted skills, and postconditions fixed, VBM-VISION reaches 98.5% C.1 success at $H{=}5$ and 98.9% C.2 success at $H{=}3$, with zero postcondition failures in both protocols. The remaining 1.5% and 1.1% failures are collision events from the shared scripted skills. Shuffling the RGB-derived coverage rows within each candidate set reduces success to 14.9% and 11.4%, respectively, confirming that the result depends on candidate-image content rather than simulator-derived coverage metadata at inference.

| Protocol | Coverage source | Succ. | Coll. | PF | Target | Select. |
|---|---|---|---|---|---|---|
| C.1, $H{=}5$ | VBM-Vision (RGB) | **98.5** | 1.5 | 0.0 | – | – |
| C.1, $H{=}5$ | Shuffled RGB | 14.9 | 1.2 | 21.1 | – | – |
| C.2, $H{=}3$ | VBM-Vision (RGB) | **98.9** | 1.1 | 0.0 | 100.0 | 90.0 |
| C.2, $H{=}3$ | Shuffled RGB | 11.4 | 0.8 | 22.5 | 33.1 | 0.6 |

Table 2. **Image-derived functional coverage.** VBM-VISION replaces simulator-derived coverage metadata at inference with a frozen RGB signature and a scene-group out-of-fold linear coverage head. The head is supervised only by predicate-visibility labels from training folds; the fixed command binding is applied after task-agnostic visual prediction. The evaluator, scripted skills, postconditions, candidate sets, and budgets are unchanged. *Shuffled RGB* permutes the predicted coverage rows among the twelve candidates within each case; its collapse verifies that the positive result depends on candidate-image content. Succ. = success (%); Coll. = collision (%); PF = postcondition failure (%); Target = target-cue rate (%); Select. = target − wrong-command cue rate. Each row aggregates 4,500 evaluator episodes.

## 6. Discussion and Conclusion

Seeing What Matters reframes spatial reasoning for simulated embodied manipulation as actionability: can the agent see what matters for the current command before a shared skill executes? The evidence is strongest in the same-object functional-OOD split, where generic object views and wrong-command affordances are insufficient—a distinction that separates the benchmark from scene-level spatial QA and category-level affordance recognition. The Spatial–Functional Belief Ledger and diagnostic reference pipeline show that command-bound evidence selection converts spatial-functional belief into closed-loop task success under equal observation budgets, while optional Action-Induced Belief Revision diagnoses recovery after postcondition mismatch.

The main empirical finding is not that one foundation model fails, but that seven strict-clean VLMs cluster within a 16.9-point band and remain 55.9 points below the diagnostic reference selector under the same primary information contract.

The benchmark is deliberately scoped: it evaluates information selection and belief revision with frozen RGB candidate views, scripted skills, and simulator-state postconditions. The diagnostic reference scorer uses semantic view names and nominal functional-part coverage—an auditable anchor, not a learned perception model or an optimal view-selection policy. The current benchmark is not a physical robot deployment, a learned dexterous controller, or a claim of household-scale generality; it extends the original Theory-of-Space probing question from scene-level spatial belief to manipulation-conditioned actionability.

VBM-VISION eliminates simulator-derived coverage metadata at inference but retains predicate-visibility labels for training the coverage head; a fully unsupervised variant remains future work.

The 500 evaluation cases come from one simulated coffee workstation; episode totals reflect method, budget, and repeat combinations over these cases, not additional domains. The Drawer/Cabinet and Microwave/Appliance interfaces provide small sanity checks, not evidence of household-scale transfer. The primary splits are close to saturation for the reference selector, which is why the shortcut controls, no-image controls, paired scene-group bootstrap, and hard $H{=}1$ stress split are important. The VLM ranking test previews all candidates simultaneously, so its gap with the sequential reference

selector is not a controlled architecture comparison.

The Drawer/Cabinet and Microwave/Appliance interfaces provide small-scale cross-domain sanity checks (180 cases each); broader evaluation with more object families, real robot execution, and learned view scorers remains future work.

The results suggest that future VLM/VLA manipulation benchmarks should evaluate not only what a model knows about a scene, but whether it can acquire the evidence that the next action requires.

**LLM Usage Statement.** The authors used a large language model for editorial feedback during manuscript preparation. All numerical results and quantitative claims were independently verified against frozen experiment records.Figures were generated from audited data or manually reviewed vector schematics; Figures 1, 3, and 4 incorporate real MuJoCo RGB exemplars.

**Acknowledgments.** We thank Guangyu Li (Hong Kong University of Science and Technology / Mohamed bin Zayed University of Artificial Intelligence) for providing the initial simulation draft that supported early development of this benchmark.

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# Seeing What Matters: Benchmarking Task-Conditioned Actionable Spatial Belief for Manipulation

Supplementary Material

## A. Supplementary Scope

The main paper carries the central claim: task-conditioned spatial-functional evidence selection improves closed-loop manipulation under a fixed scripted-skill interface. This supplement contains every distinct scientific result and audit boundary supporting the paper: the information boundary, baseline fairness contract, paired scene-group bootstrap, strict-clean seven-model evidence, charged-view costs, aggregate and task-level ablations, held-out-domain sanity checks, hard one-view stress split, logged rollout trace, raw MuJoCo grounding frames, and compact task/failure diagnostics.

We follow three rules, summarized in Table 3. First, illustrative mechanism figures and empirical evidence are separated. The main paper's hybrid schematics explain the benchmark interface; the supplement's quantitative plots and tables trace to frozen rollouts, CSV summaries, or packaged manifests. Second, strict-clean, primary-run, and ablation-run records remain explicitly separated. Third, a value set is shown once in its most auditable form; complementary plots are retained only when they expose a relationship not legible from an exact table.

## B. Benchmark Construction and Information Boundary

The benchmark is a task-conditioned analogue of spatial-belief probing. Instead of asking whether a model can build a complete scene-level cognitive map, each episode asks whether a policy acquires the spatial and functional evidence needed by the current operation. The five primary coffee-task families are TAMP, INSERT, PLACE, SERVE, and GRIND. Each family contains 50 scene groups and two variants, giving 100 variants per task and 500 variants per protocol before method and inference-repeat expansion.

Each episode contains a hidden simulator state, a command, a fixed set of candidate views, and an observation budget. The agent receives the command, acquired RGB frames, view metadata, and its own belief state. It does not receive hidden simulator predicates, evaluator labels, unselected endpoint renders, contact logs, or postcondition truth before acting. The same scripted skill library is used across methods, so differences in closed-loop success come from information selection and belief revision rather than from changing the low-level controller.

## C. Raw MuJoCo Rendering Examples

The main-paper figures use hybrid schematics to explain the task-conditioned evidence mechanism. For auditability, Figs. 6 and 7 provide raw MuJoCo RGB frames from the frozen simulator interface. These filmstrips are not intended as photorealistic renderings; they document the candidate views, task labels, and held-out domains used by the benchmark. The released artifact additionally includes short qualitative GIF/MP4 demos with raw rollout JSON; quantitative claims in the paper are computed from frozen CSV postconditions, not from these videos.

## D. Protocols, Metrics, and Baseline Fairness

**Protocols.** C.1 is the closed-loop actionability protocol. Its primary budget is $H = 5$, and it evaluates whether selected observations support successful execution of the shared scripted skill. C.2 is the same-object functional-OOD protocol. Its primary budget is $H = 3$, and it evaluates whether methods observe the command-relevant functional cue rather than a plausible cue for a different command. Recovery experiments evaluate postcondition mismatch handling after an initially stale or contradicted belief.

**Metrics.** Task success is the percentage of episodes whose shared scripted skill satisfies the hidden simulator-state postcondition. Collision rate and postcondition-failure rate are reported separately. Average charged views count all acquired views, including recovery observations. Primitive steps count scripted skill steps. Unknown/abstain is tracked separately from failed execution. In C.2, target-functional hit measures whether selected views expose the functional cue required by the current command, wrong-command hit measures whether selected views expose a cue for another command, and command selectivity is their difference in percentage points.

**Shortcut controls.** Table 4 states the common access contract. Table 5 is a same-object shortcut stress test. Category shortcut, command-agnostic affordance, command-only/no-image, no-functional-fields, and no-target-conditioning controls all use the same hidden postcondition evaluator as the full method. Their failures rule out several common leakage explanations: object category alone, plausible but wrong-command affordance evidence, command templates without visual evidence, pose/category evidence without functional predicates, and target selection without command binding.

| Retained evidence | Why it stays in the supplement |
|---|---|
| Raw MuJoCo filmstrips | Ground the benchmark in actual simulator RGB frames, especially because the main figures are hybrid schematics. |
| Baseline fairness and metrics contract | Makes clear which methods see commands, ledgers, functional fields, AIBR, or privileged labels. |
| Sanity-checks and ablation tables | Detailed control, component-removal, and ablation tables supporting the main-paper text paragraphs. |
| Paired scene-group bootstrap | Checks that primary gains do not come from repeated answers on the same case family. |
| Strict-clean VLM protocols | Documents the seven-model 1,500-case strict-clean gate and results (Tables 7–8 and Table 9); earlier 500-row records are archived in the release artifact. |
| Charged-view and ablation evidence | Completes the cost and component analyses promised by the main paper without substituting primary- and ablation-run values. |
| Budget-curve and diagnostic visualizations | Budget-sweep, functional-OOD evidence, and recovery stress-test figures complementing the main results table. |
| Held-out Drawer/Microwave sanity check | Addresses the single-coffee-domain concern with small but auditable cross-domain evidence. |
| Hard $H = 1$ stress split and logged trace | Shows residual difficulty under tight budgets and gives an auditable single-case score/select/verify trace. |
| Per-task and failure diagnostics | Checks whether aggregate results hide one easy task family or unsafe failure mode. |
| Reproducibility record | SHA-256 manifests, package validation, chart verification, and provenance chain for regeneration. |

Table 3. **Supplement roadmap.** Every distinct scientific result and audit boundary is retained once; duplicate presentations are removed.




Figure 6. **Primary five-task visual domain.** Raw MuJoCo/Panda RGB exemplars from the coffee workstation show that tamping, insertion, placement, serving, and grinding require different command-relevant functional evidence even when the object family is shared.

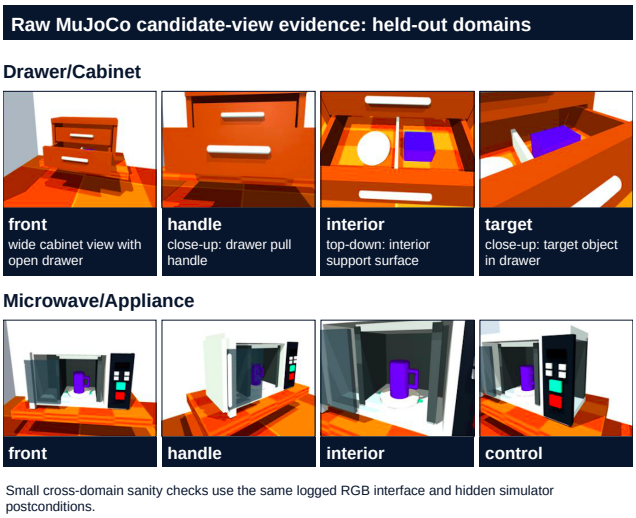



Figure 7. **Raw MuJoCo candidate-view examples for held-out domains.** Drawer/Cabinet and Microwave/Appliance use the same logged RGB candidate-view interface and hidden simulator-state postcondition evaluator as the coffee benchmark.

## E. Paired Scene-Group Bootstrap

Table 6 reports paired scene-group bootstrap intervals using the regenerated episode-level records. For each frozen case, identified by task, group, and variant, we average the three inference repeats for each method. We then resample the 500 paired case clusters with replacement for 10,000 bootstrap replicates. This directly addresses whether the primary aggregate gains depend on repeated answers from the same case. The main conclusion is stable: the View Belief Map with optional AIBR strongly improves over History active in C.1 and over command-agnostic affordance in C.2. The optional AIBR versus View Belief Map rows are small, as expected; they show that the functional-OOD gain is driven mainly by command-conditioned view selection rather than by postcondition repair.

| Method | Cmd. | Ledger | Functional fields | AIBR | Privileged | Purpose |
|---|---|---|---|---|---|---|
| RANDOM | ✗ | ✗ | ✗ | ✗ | ✗ | Chance valid-view policy under the same view budget. |
| FIXED SCAN | ✗ | ✗ | ✗ | ✗ | ✗ | Deterministic camera-order policy; tests whether coverage alone solves the benchmark. |
| POSE COVERAGE | ✗ | ✗ | ✗ | ✗ | ✗ | Uses candidate-pose geometry only; no hidden visibility or task-state labels. |
| HISTORY ACTIVE | ✓ | ✗ | ✗ | ✗ | ✗ | Active view selection from observation history, without structured missing-predicate bookkeeping. |
| TARGET LEDGER | ✓ | ✓ | Partial | ✗ | ✗ | Stores target predicates but does not score views by missing functional evidence. |
| VIEW BELIEF MAP | ✓ | ✓ | ✓ | ✗ | ✗ | Scores candidate views by expected evidence gain over missing target-relevant predicates. |
| OURS+AIBR | ✓ | ✓ | ✓ | ✓ | ✗ | Optional local repair after postcondition mismatch; not the main source of primary-split gains. |
| PRIV. DIAGNOSTIC | ✓ | ✓ | ✓ | Optional | ✓ | Privileged diagnostic selector; not a mathematical upper bound and excluded from deployable-method ranking. |

Table 4. **Baseline fairness contract.** All non-privileged methods operate under the same candidate views, budgets, scripted skills, and hidden evaluator. "Cmd." denotes whether the command is available to the selection rule; "Privileged" denotes access to hidden state, hidden visibility, or postcondition truth before action.

| | *Functional-OOD metrics (%)* | | | | |
|---|---|---|---|---|---|
| Control | Succ. ↑ | Target ↑ | Wrong | Select. ↑ | What the control rules out |
| *Deployable method* | | | | | |
| FULL METHOD | **98.9** | **100.0** | 30.0 | **70.0** | The deployable policy succeeds while selecting command-relevant functional evidence much more often than wrong-command evidence. |
| *Shortcut and leakage controls* | | | | | |
| CATEGORY SHORTCUT | 3.7 | 26.7 | 26.7 | 0.0 | Seeing the correct object category is insufficient when the same object supports multiple commands. |
| CMD-AGNOSTIC AFFORD. | 19.7 | 46.7 | 73.3 | −26.7 | Generic affordance viewing can expose plausible but wrong-command parts and still fail the postcondition. |
| COMMAND-ONLY/NO-IMAGE | 0.0 | 0.0 | 0.0 | 0.0 | Removing all RGB evidence eliminates visual verification, ruling out command-template execution. |
| *Component removal* | | | | | |
| w/o functional fields | 0.0 | 16.7 | 23.3 | −6.7 | The score cannot be explained by pose-only or object-only evidence. |
| w/o target conditioning | 3.5 | 26.7 | 26.7 | 0.0 | Removing command-conditioned missing-predicate selection destroys the useful-view advantage. |

Table 5. **Sanity checks for information leakage and shortcut behavior.** All rows use the same-object functional-OOD hidden evaluator and 1,500 episodes per method. Target cue and Wrong cue are separately measured functional-cue hits; Select. is their difference. The command-only/no-image row reruns the primary cases with all RGB evidence removed. The controls verify that the main success is not explained by category recognition, command-agnostic affordance viewing, pose-only evidence, command templates, or access to hidden postconditions.

| Protocol | Paired comparison | Cases | Mean diff. | 95% CI / interpretation |
|---|---|---|---|---|
| C.1 | OURS+AIBR – HISTORY ACTIVE | 500 | 45.9 | [41.9, 49.8], large gain. |
| C.1 | OURS+AIBR – VIEW BELIEF MAP | 500 | 0.9 | [-0.1, 1.8], small AIBR increment. |
| C.2 | OURS+AIBR – CMD-AGNOSTIC AFFORD. | 500 | 79.1 | [75.7, 82.6], large gain. |
| C.2 | OURS+AIBR – VIEW BELIEF MAP | 500 | 0.1 | [-0.7, 0.8], small AIBR increment. |
| C.2 | OURS+AIBR – COMMAND ONLY / NO IMAGE | 500 | 98.9 | [98.3, 99.4], rules out command-only execution. |

Table 6. **Paired scene-group bootstrap over frozen cases.** Differences are success-rate percentage points. We average the three inference repeats for each frozen case, then resample the 500 paired case clusters (task, group, variant) with replacement for 10,000 bootstrap replicates.

# F. Foundation-Model and Cross-Domain Review Checks

## F.1. Strict-clean seven-model view-scoring evidence

Each case presents the command and all twelve frozen RGB candidates, then asks the model to return an ordered set

of exactly three unique view IDs. This is a full-preview ranking diagnostic, not an equal-information comparison with sequential acquisition. A promoted row must pass endpoint/HTTP status, strict JSON, exact unique $H = 3$, candidate membership, complete evaluator record, exact case order, and no-fallback checks for all 1,500 cases. The unchanged hidden simulator-state evaluator scores the selected views. Earlier 500-row endpoint records are archived in the release artifact; the strict-clean protocol supersedes them. Exact InternVL3-38B and Molmo-72B endpoints were recorded as unavailable under the same endpoint protocol; no numerical rows are invented or substituted for them.

Per-task strict-clean results are reported in Table 9. Tables 7–8 retain the admission contract and full cue diagnostics.

**VBM-Vision method details.** We compute a frozen CLIP-B/32 image-text signature for each of the 18,000 candidate RGB records and fit one class-balanced logistic-regression head per functional predicate. We use five scene-group folds (150 groups total; 120 training and 30 held-out groups per fold), fixed $C$=1.0, the `liblinear` solver, and no evaluator-success-based hyperparameter tuning. All reported coverage scores are strictly out of fold. The coverage head is trained using nominal predicate-visibility labels from the training folds, but at inference it receives only frozen RGB-derived features and candidate-relative normalizations. It does not receive task or command identity, view or camera identifiers, simulator state, success labels, evaluator outputs, or the nominal view-to-predicate coverage mapping. The command is bound to predicted predicates only after this task-agnostic visual stage.

**Held-out Drawer/Microwave cross-domain sanity check.** To probe whether the actionability formulation is tied only to the coffee workstation, we add two small held-out MuJoCo domains: Drawer/Cabinet and Microwave/Appliance. Each domain contains 180 cases and 1,800 rendered candidate RGB images. Table 10 summarizes the task types and hidden postcondition checks. The same transparent selector and hidden postcondition interface are reused without domain-specific retuning and without claiming broad household-scale generality.

Table 11 reports the final equal-budget $H = 2$ rerun used for submission. View Belief Map+Action-Induced Belief Revision transfers above random view selection on both domains, but does not saturate them. Drawer/Cabinet is repair-sensitive: Action-Induced Belief Revision raises the transparent reference selector from 50.0% to 66.7%, corresponding to 30 additional recovered cases out of 180. Microwave/Appliance remains at 66.7% for the ledger-based rows at this small scale. Table 12 audits case-level equality and selected-view traces rather than relying only on aggregate success. The lower absolute success relative to the coffee primary split reflects new articulated-object predicates, task templates, and view names for which the reference rules were not rewritten.

Each held-out domain uses three task families, compared with five in the primary coffee benchmark, to keep the cross-domain check tractable at 180 cases per domain. Drawer/Cabinet covers opening, retrieval, and placement; Microwave/Appliance covers opening, placement, and activation. These checks probe transfer of the evaluation interface across object families rather than matching the primary domain's task coverage.

## G. Hard Budget and Logged-Trace Evidence

**Hard one-view stress split.** The main protocols use $H = 5$ for C.1 and $H = 3$ for C.2, where the transparent reference selector approaches saturation. Table 13 and Fig. 10 expose a harder diagnostic by filtering the frozen episode logs to $H = 1$. This is not a new simulator run or a manually edited table; it is a re-summarization of the logged C.1 and C.2 budget-sweep episodes. The one-view constraint makes the task substantially harder: VBM+AIBR reaches 19.0% in C.1 and 58.6% in C.2, far below the primary near-ceiling results. Under $H = 1$, the privileged diagnostic reaches 48.0% in C.1 while VBM+AIBR reaches 19.0%, indicating a substantial headroom for future learned view selectors within the same information contract.

**Logged rollout trace.** Figure 11 replaces a purely conceptual selection illustration with a logged functional-OOD trace. The selected case is a successful GRIND intervention episode with a TAMP distractor under $H = 1$. The panel shows the actual candidate-view scores stored in the JSONL trace, the selected view, observed fields, target functional predicates, and postcondition pass count. It is therefore a compact audit trail for the central mechanism: the selector succeeds by choosing the view that exposes the command-relevant functional chain.

## H. Charged-View Cost and Ablation Detail

**Actual charged views across the budget sweep.** Most methods use exactly the initial budget. Recovery-capable and privileged diagnostic rows can use additional charged observations after mismatch at $H = 1$; the full budget curves are shown in Fig. 12, and the nontrivial reference rows are listed exactly in Table 14. These costs are distinct from success rates.

**Ablation-run values.** Table 15 reports the primary functional-OOD ablation. Table 16 reports the frozen C.1 ablation aggregate. Its full-method value (97.60%) belongs to the ablation run and must not be substituted for the 98.8%

| Gate stage | Requirement | Retained evidence | Promoted condition |
|---|---|---|---|
| Dry-run probe | Five frozen probes return strict JSON with exactly three unique candidate IDs. | Provider smoke/probe logs; a failed endpoint is not promoted. | 5/5 |
| Strict prompt | Exactly three unique IDs, all among the twelve frozen candidates. | Prompt template and model-specific run manifest. | 1,500/1,500 |
| Frozen manifest | Same order, 300 cases/task, twelve RGB candidates/case. | Strict manifest and render audit; 18,000 hashed RGB images. | 1,500/1,500 |
| Endpoint retry | Retry only failed endpoint rows; endpoint and format repair stay separate. | Retry logs and manifests retained without overwriting originals. | Clean rows only |
| Raw responses | One retained raw response per promoted row. | `raw_responses.jsonl`; failed/fallback rows remain separate. | 1,500/1,500 |
| Parsed selections | Strict JSON only; no regex/fallback extraction. | Parsed selections and strict-gate JSON. | 1,500/1,500 |
| Episode results | Unchanged hidden postcondition evaluator. | Episode results plus task/overall summaries. | 1,500/1,500 |
| Combined provenance | Combine only clean original/retry/repair rows, preserving order. | Provenance audit retains the original case order. | Exact order |
| SHA256/checks | Hash-check manifests and RGB records. | Manifest/render audits and package SHA256 records. | Pass |

Table 7. **Strict-clean VLM promotion checklist.** This is the archived admission contract, not a claim that the present document rebuild reran remote model responses.

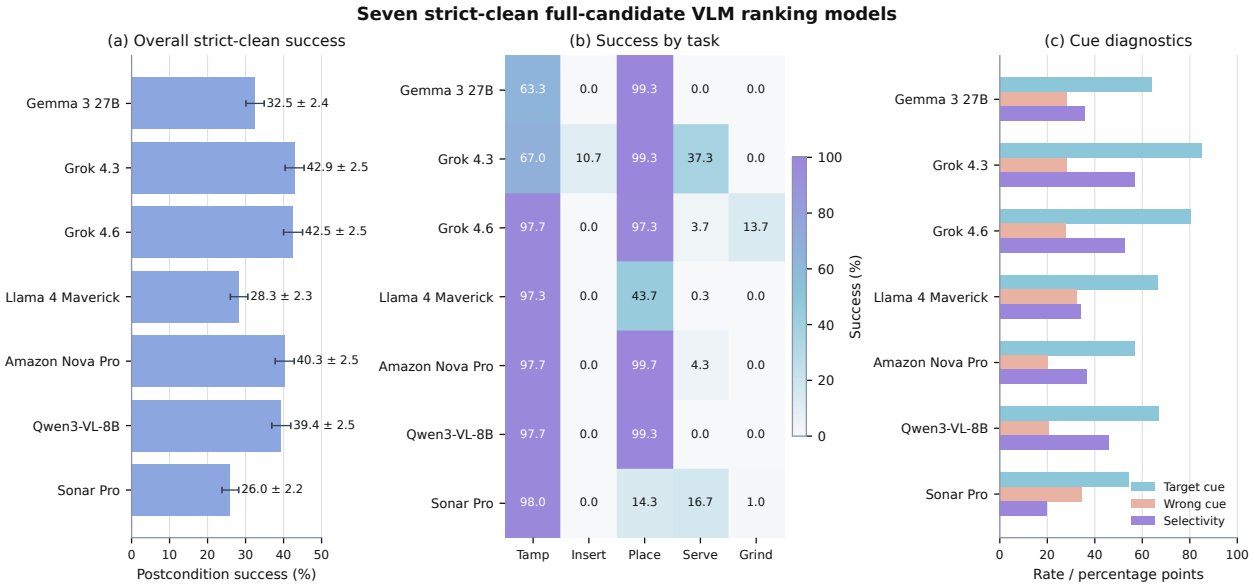



Figure 8. **Seven-model strict-clean comparison.** (a) Overall postcondition success with archived normal-approximation 95% half-width. (b) Per-task success. (c) Target-cue, wrong-cue, and selectivity diagnostics. Generated only from the archived strict-clean CSVs; it does not equate the VLM full-preview contract with sequential acquisition.

| Model | Success ↑ | Target ↑ | Wrong ↓ | Select. ↑ | Coll. ↓ | Post-fail ↓ | $n$ |
|---|---|---|---|---|---|---|---|
| Gemma 3 27B | 32.5 ± 2.4 | 63.8 | 28.1 | 35.6 | 3.7 | 63.7 | 1,500 |
| Grok 4.3 | 42.9 ± 2.5 | **85.2** | 28.2 | **57.0** | 2.6 | **54.5** | 1,500 |
| Grok 4.6 | 42.5 ± 2.5 | 80.2 | 27.8 | 52.4 | **2.4** | 55.1 | 1,500 |
| Llama 4 Maverick | 28.3 ± 2.3 | 66.3 | 32.3 | 34.0 | 3.1 | 68.6 | 1,500 |
| Amazon Nova Pro | 40.3 ± 2.5 | 57.0 | **20.3** | 36.7 | 3.5 | 56.1 | 1,500 |
| Qwen3-VL-8B | 39.4 ± 2.5 | 66.7 | 20.7 | 45.9 | 3.2 | 57.4 | 1,500 |
| Sonar Pro | 26.0 ± 2.2 | 54.3 | 34.4 | 19.9 | 3.5 | 70.5 | 1,500 |

Table 8. **Verified strict-clean foundation-model view scoring.** Values are percentages except $n$; ± is the archived normal-approximation 95% interval half-width. Every row averages 3.0 selected views and 2.6 primitive steps. Target/wrong cue and selectivity are evaluator diagnostics; success is the hidden postcondition outcome. Diagnostics were rounded independently.

primary-run value. Table 17 exposes the corresponding task structure for C.1 and C.2.

# I. Compact Diagnostic Figures

The following figures test whether the main aggregate hides task-family or failure-mode artifacts. They complement, rather than repeat, the exact cost and ablation tables above.

Figures 15–18 respectively expose task-level success, nonexclusive failure modes, recovery by task, and diagnostic QA accuracy; none is used as a substitute for simulator postcondition success.

# J. Data Audit and Provenance

The regenerated figures use the frozen CSVs stored in `data/`. During audit, several transcription differences were found and corrected in the C.2 summaries, including the no-functional-fields row. In the final re-

| Strict-clean VLM | Tamp | Insert | Place | Serve | Grind | Overall |
|---|---|---|---|---|---|---|
| Gemma 3 27B | 63.3 | 0.0 | 99.3 | 0.0 | 0.0 | 32.5±2.4 |
| Grok 4.3 | 67.0 | 10.7 | 99.3 | 37.3 | 0.0 | 42.9±2.5 |
| Grok 4.6 | 97.7 | 0.0 | 97.3 | 3.7 | 13.7 | 42.5±2.5 |
| Llama 4 Maverick | 97.3 | 0.0 | 43.7 | 0.3 | 0.0 | 28.3±2.3 |
| Amazon Nova Pro | 97.7 | 0.0 | 99.7 | 4.3 | 0.0 | 40.3±2.5 |
| Qwen3-VL-8B | 97.7 | 0.0 | 99.3 | 0.0 | 0.0 | 39.4±2.5 |
| Sonar Pro | 98.0 | 0.0 | 14.3 | 16.7 | 1.0 | 26.0±2.2 |

Table 9. **Strict-clean VLM postcondition success by task.** Each model is evaluated on 1,500 cases (300 per task) under the same frozen candidate-view and hidden-postcondition contract. Overall includes normal-approximation 95% intervals. Task-level asymmetry is pronounced: models that approach perfect success on Tamp or Place can remain near zero on Insert, Serve, or Grind. Admission gate and complete diagnostics appear in Tables 7 and 8.

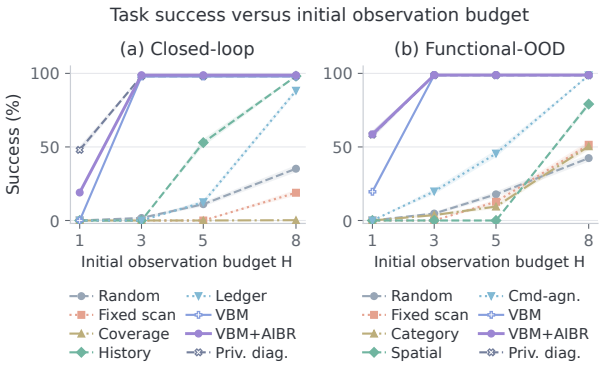



Figure 9. **Task success versus initial observation budget.** Lines connect only the observed budgets $H \in \{1, 3, 5, 8\}$. Recovery methods may use extra charged recovery observations after the initial budget, which are analyzed separately in the supplementary charged-view curves.

generated figures and tables, no-functional-fields success is 0.0%; the 16.7 value refers to target-cue observation. The source package contains regenerated CSVs and verification records under `data/`, MuJoCo run and validation scripts under `scripts/`, and review records including `REPRODUCIBILITY.md`, `DATA_CARD.md`, `ASSET_ATTRIBUTION.md`, `COMPUTE_REPORT.md`, and `APPENDIX_QA_CHECKLIST.md`. The files `DATA_DIFFERENCES.md`, `figure_catalog.csv`, `plotted_values.csv`, and `verification.json` trace the plotted numbers and table values back to source summaries.

**Asset and figure provenance.** The two mechanism figures in the main paper are hybrid schematics depicting the coffee workstation, Drawer/Cabinet, and Microwave/Appliance with abstract modules and real MuJoCo RGB exemplars. They explain the benchmark interface and command-dependent evidence requirements; their annotations are not learned attention maps, segmentation masks, recorded selections, or quantitative outcome evidence. Quantitative plots in this supplement are generated from frozen CSV summaries; the mapping between plotted figures and source values is recorded in `figure_catalog.csv` and `plotted_values.csv`. Raw MuJoCo filmstrips document the simulator visual domain, while task success and failure claims are based on simulator-state postconditions.

## K. Reproducibility Record

The review package is organized so that the main claims can be regenerated from frozen records rather than from manually edited numbers. All experiment stages retain schedules, checkpoints, raw traces, CSVs, generated tables, run logs, and SHA-256 manifests. The package includes a reproducibility guide, data card, asset-attribution note, CVPR-style compute report, and appendix QA checklist. The two main MuJoCo output packages have SHA-256 prefixes `045931cb445c` and `63011d469d`. The canonical functional-OOD run validates with `ok: true`, 55,500 episode rows, 55,500 schedule rows, and 55,500 trace lines. The final chart package verifies 1,744 plotted/table source values after adding strict-clean VLM overall and per-task rows; it stores the plotted CSVs, figure catalog, and data-difference audit alongside the source. The supplement includes seven verified 1,500-case strict-clean VLM rows, public RGB dataset package hashes, a hard $H=1$ stress summary derived from frozen episodes, and a logged rollout trace figure generated from candidate-view scores and postcondition records.

## L. Limitations of the Supplementary Evidence

The supplementary evidence strengthens auditability but does not over-broaden the paper's claim. The experiments evaluate simulated information selection and belief revision with frozen RGB candidate views, shared scripted skills, and hidden simulator-state postconditions. The direct VLM stress check and Drawer/Microwave sanity check address two reviewer concerns–whether a prompted foundation model can be evaluated under the same contract and whether the formulation is confined to one coffee object family–but they do not demonstrate physical robot deployment, learned dexterous control, or household-scale generality. The main bootstrap table reports paired scene-group intervals over frozen cases, while budget-curve bands remain descriptive episode-level intervals. A future version should add independent environment seeds and real-robot trials where possible. Finally, diagnostic VLM QA remains a support check: it indicates whether selected images contain answerable spatial-functional information, but the main performance claim is closed-loop simulator postcondition success.

| Domain | Task types | Hidden postcondition examples |
|---|---|---|
| Drawer/Cabinet | open drawer; retrieve object; place object | drawer openness; handle access; target visibility/reachability after opening; placement on commanded support within distance threshold. |
| Microwave/Appliance | open door; place item; activate control | door open/closed state; item inside or aligned with cavity/support; commanded control/button state; collision and wrong-control failures tracked separately. |

Table 10. **Held-out domain protocol details.** Each domain contains 180 cases and 1,800 rendered RGB candidate views. Drawer/Cabinet and Microwave/Appliance use articulated-object states and postcondition checks that differ from the coffee workstation. The numbers are intentionally small sanity checks, not a claim of household-scale domain coverage.

| Method | Drawer | Microwave | Avg. |
|---|---|---|---|
| Random view | 6.1 | 8.3 | 7.2 |
| Fixed scan | 0.0 | 0.0 | 0.0 |
| Pose coverage | 0.0 | 0.0 | 0.0 |
| Category shortcut | 0.0 | 0.0 | 0.0 |
| Cmd.-agnostic afford. | 0.0 | 0.0 | 0.0 |
| VBM | 50.0 | 66.7 | 58.4 |
| VBM+AIBR | 66.7 | 66.7 | 66.7 |
| PRIV. DIAGNOSTIC | 50.0 | 66.7 | 58.4 |

Table 11. **Held-out cross-domain sanity check under equal $H = 2$ budget.** We add two small non-coffee MuJoCo domains, Drawer/Cabinet and Microwave/Appliance, each with 180 cases and 1,800 rendered RGB candidate views. Drawer exposes a repair-sensitive setting: AIBR recovers 30 additional cases over VBM. Microwave has identical ledger-method success sets at this scale, but the raw audit records different VBM and privileged selected-view traces. These results support the paper's claim as a stress sanity check rather than a household-scale generality result.

| Domain / comparison | Success sets equal? | Selected-view traces equal? | A-only | B-only |
|---|---|---|---|---|
| Drawer/Cabinet VBM vs. VBM+AIBR | No | Yes | 0 | 30 |
| Drawer/Cabinet VBM vs. PRIV. DIAGNOSTIC | Yes | Yes | 0 | 0 |
| Microwave/Appliance VBM vs. VBM+AIBR | Yes | Yes | 0 | 0 |
| Microwave/Appliance VBM vs. PRIV. DIAGNOSTIC | Yes | No | 60 | 60 |

Table 12. **Raw case-set and selected-view audit for the held-out $H = 2$ stress check.** Each row compares two method outputs over the same 180 frozen cases. Drawer shows a genuine repair gain: VBM+AIBR succeeds on 30 cases that VBM fails, while keeping the same selected-view trace before repair. Microwave has the same success set for the ledger-based rows, but VBM and the privileged diagnostic choose different view traces for 60 cases in each direction, indicating that identical 66.7% success is not produced by copied method rows. A/B follow method order; the nonzero rows count success-set (Drawer) and selected-view-trace (Microwave) differences, respectively.

| Protocol | Method | $H = 1$ | $H = 3$ | $H = 5$ | $H = 8$ |
|---|---|---|---|---|---|
| C.1 | VBM | 1.000 | 3.000 | 5.000 | 8.000 |
| C.1 | VBM+AIBR | 1.287 | 3.000 | 5.000 | 8.000 |
| C.1 | Priv. diagnostic | 1.960 | 3.000 | 5.000 | 8.000 |
| C.2 | VBM | 1.000 | 3.000 | 5.000 | 8.000 |
| C.2 | VBM+AIBR | 1.389 | 3.000 | 5.000 | 8.000 |
| C.2 | Priv. diagnostic | 1.775 | 3.000 | 5.000 | 8.000 |

Table 14. **Charged-view values for recovery-capable/reference rows.** Other rows in the frozen C.1/C.2 budget summaries use exactly $H$ charged views.

| Variant | Succ. ↑ | Target ↑ | Wrong | Select. ↑ | Coll. ↓ | PostFail ↓ |
|---|---|---|---|---|---|---|
| **Full method** | **98.9** | **100.0** | 30.0 | **70.0** | **1.1** | **0.0** |
| w/o AIBR | 98.9 | 100.0 | 30.0 | 70.0 | 1.1 | **0.0** |
| w/o command binding | 19.7 | 46.7 | 73.3 | −26.7 | 3.9 | 76.3 |
| w/o functional fields | 0.0 | 16.7 | 23.3 | −6.7 | 3.5 | 96.5 |
| w/o target conditioning | 3.5 | 26.7 | 26.7 | 0.0 | 5.7 | 90.7 |

Table 15. **Ablation under functional-OOD.** Command binding and functional fields are the decisive components; without them, success collapses even when the same object category is visible. The 16.7 value for the no-functional variant is target-cue observation, not task success. **Bold**: best; underline: second-best.

| C.1 variant | Success | Collision | Post-fail | Views | $n$ |
|---|---|---|---|---|---|
| Full method | 97.60 | 2.40 | 0.00 | 5.0 | 1,500 |
| w/o AIBR | 98.13 | 1.87 | 0.00 | 5.0 | 1,500 |
| w/o belief memory | 47.33 | 2.53 | 50.13 | 5.0 | 1,500 |
| w/o functional fields | 99.00 | 1.00 | 0.00 | 5.0 | 1,500 |
| w/o target conditioning | 0.00 | 5.00 | 95.00 | 5.0 | 1,500 |

Table 16. **Frozen C.1 ablation aggregate.** Values are percentages except views and $n$. This ablation-run summary is kept separate from the primary result.

## M. Evidence Provenance and Release Chain

The evidence package keeps complementary protocols explicit: the main paper's seven-model strict-clean experiment is supported by Tables 7–8, Fig. 8, and Table 9. Quantitative Drawer/Microwave results remain the small interface sanity check in Tables 10–12. Charged-view and ablation evidence appears in Tables 14–17 and Fig. 12. Content already present in the prior supplement is referenced in place rather than copied.

**Evaluator API.** The extension chain is

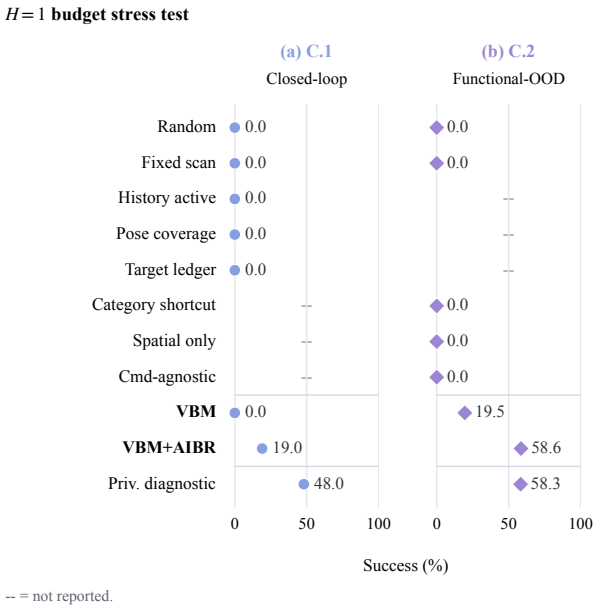



Figure 10. **Hard one-view budget stress test.** The figure is generated from the frozen episode logs by filtering to $H = 1$. It exposes residual difficulty that is obscured by the primary $H = 5$ C.1 and $H = 3$ C.2 settings. Missing protocol/method combinations are left unconnected rather than imputed. Uncertainty for reported rows is provided in Table 13.

| Method | C.1 hard H=1 | C.2 hard H=1 |
|---|---|---|
| Random | $0.0 \pm 0.0$ | $0.0 \pm 0.0$ |
| Fixed scan | $0.0 \pm 0.0$ | $0.0 \pm 0.0$ |
| History active | $0.0 \pm 0.0$ | – |
| Pose coverage | $0.0 \pm 0.0$ | – |
| Target ledger | $0.0 \pm 0.0$ | – |
| Category shortcut | – | $0.0 \pm 0.0$ |
| Spatial only | – | $0.0 \pm 0.0$ |
| Cmd-agnostic | – | $0.0 \pm 0.0$ |
| VBM | $0.0 \pm 0.0$ | $19.5 \pm 2.0$ |
| VBM+AIBR | $19.0 \pm 2.0$ | $58.6 \pm 2.5$ |
| Priv. diagnostic | $48.0 \pm 2.5$ | $58.3 \pm 2.5$ |

Table 13. **Hard budget stress test.** Success rates at an initial one-view budget, re-summarized from the frozen episode logs. This split exposes saturation hidden by the main H=3/H=5 settings and should be read as a stress diagnostic rather than a replacement for the main protocol. Here $n = 1{,}500$ per cell; $\pm$ is the half-width of a normal-approximation 95% binomial CI; and an en dash denotes a combination absent from the frozen logs. At observed zero, the approximation yields $0.0 \pm 0.0$ and does not imply zero population uncertainty.

`manifest.csv` → `selected_views.csv` → `episode_results.csv` → `strict_gate.json` → summary tables.

A new model uses the same command-plus-candidate-view contract, preserves raw responses and parsed selections,

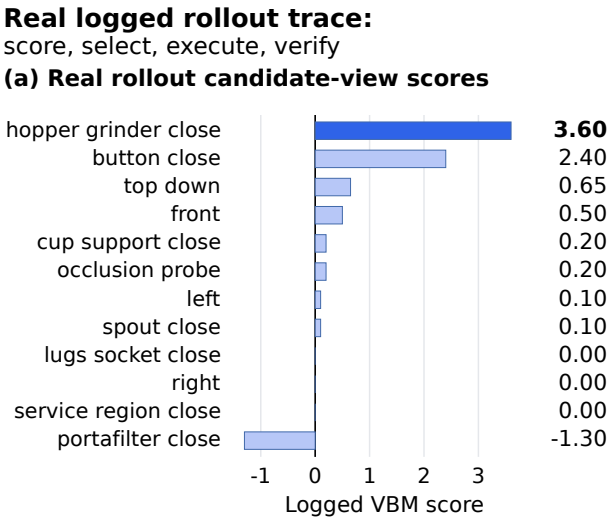



Figure 11. **Real logged rollout trace.** Candidate-view scores, selected view, observed fields, and postcondition outcome are read from the frozen trace record rather than drawn as a storyboard. This is one representative successful trace; the hard $H = 1$ aggregate success rates are reported in Table 13. The panel uses the logged score and postcondition records; RGB thumbnails from the frozen image directory are included in the release artifact.

runs the unchanged postcondition evaluator, and passes the strict gate. A new domain additionally requires a simulator scene, Panda-compatible scripted skills, a hidden postcondition evaluator, a candidate-view renderer, and a frozen manifest with SHA256 records before a quantitative row is

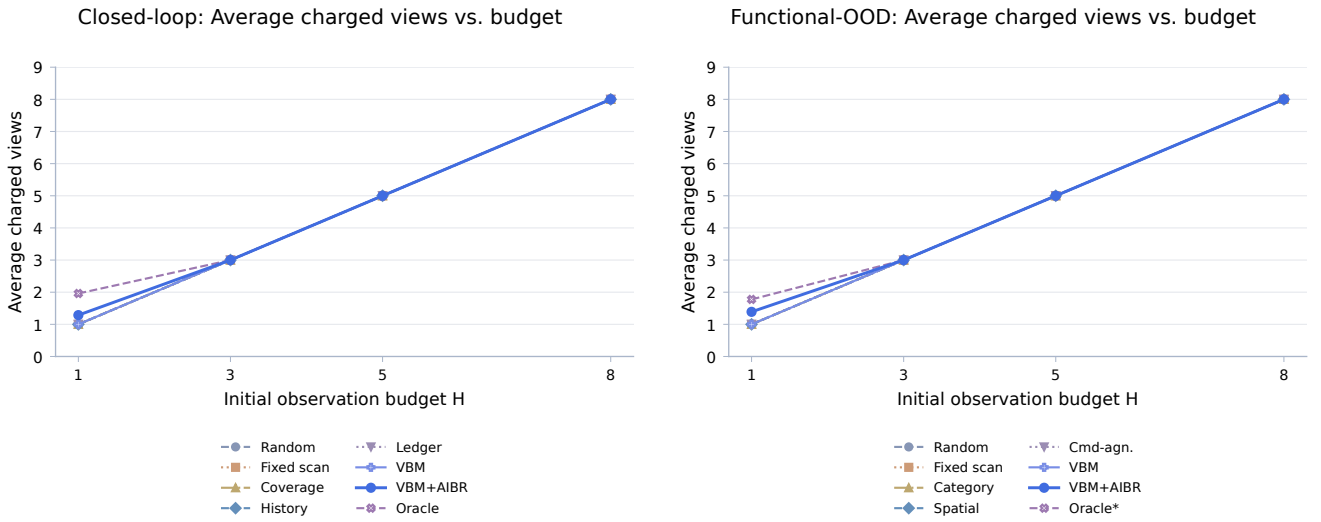



Figure 12. **Average charged views versus initial observation budget.** Left: C.1 closed-loop actionability. Right: C.2 functional-OOD. Recovery observations are charged in addition to the initial selection budget; lines connect only observed budgets $H \in \{1, 3, 5, 8\}$.

| Protocol | Variant | Tamp | Insert | Place | Serve | Grind | Average |
|---|---|---|---|---|---|---|---|
| C.1 | Full | 97.0 | 98.3 | 96.7 | 98.0 | 98.0 | 97.6 |
| C.1 | w/o AIBR | 96.7 | 98.7 | 98.3 | 98.3 | 98.7 | 98.1 |
| C.1 | w/o belief memory | 0.0 | 39.3 | 2.0 | 97.3 | 98.0 | 47.3 |
| C.1 | w/o functional fields | 99.3 | 99.3 | 98.7 | 98.7 | 99.0 | 99.0 |
| C.1 | w/o target conditioning | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| C.2 | Full | 99.0 | 99.0 | 99.0 | 98.7 | 99.0 | 98.9 |
| C.2 | w/o AIBR | 98.3 | 98.7 | 99.0 | 98.3 | 100.0 | 98.9 |
| C.2 | w/o command binding | 0.0 | 0.0 | 0.0 | 98.7 | 0.0 | 19.7 |
| C.2 | w/o functional fields | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| C.2 | w/o target conditioning | 0.0 | 0.0 | 17.7 | 0.0 | 0.0 | 3.5 |

Table 17. **Task-level ablation success (%).** Reconstructed directly from the frozen task CSVs; the matrix shows whether an aggregate difference is concentrated in one task.

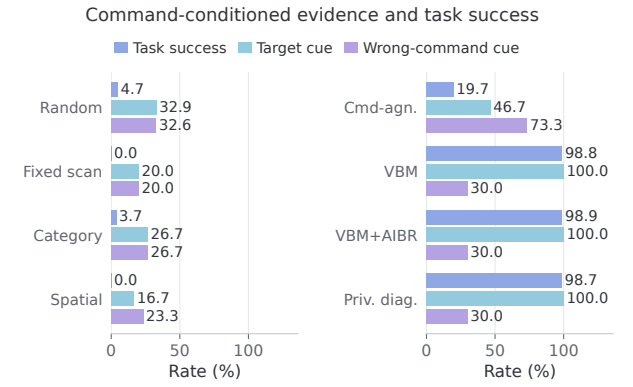



Figure 13. **Functional evidence must be command-bound.** Target-functional cue observation tracks task success, whereas wrong-command cue observation does not. Target and wrong-command cue rates are separate measurements and need not sum to 100%.

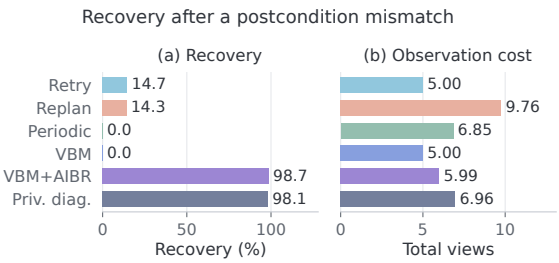



Figure 14. **Recovery stress test for postcondition mismatch.** Optional Action-Induced Belief Revision repairs stale belief fields locally and restores 98.7% of mismatch episodes in the constructed mismatch setting. The right panel reports total charged views including initial observations, not only extra recovery views.

promoted.

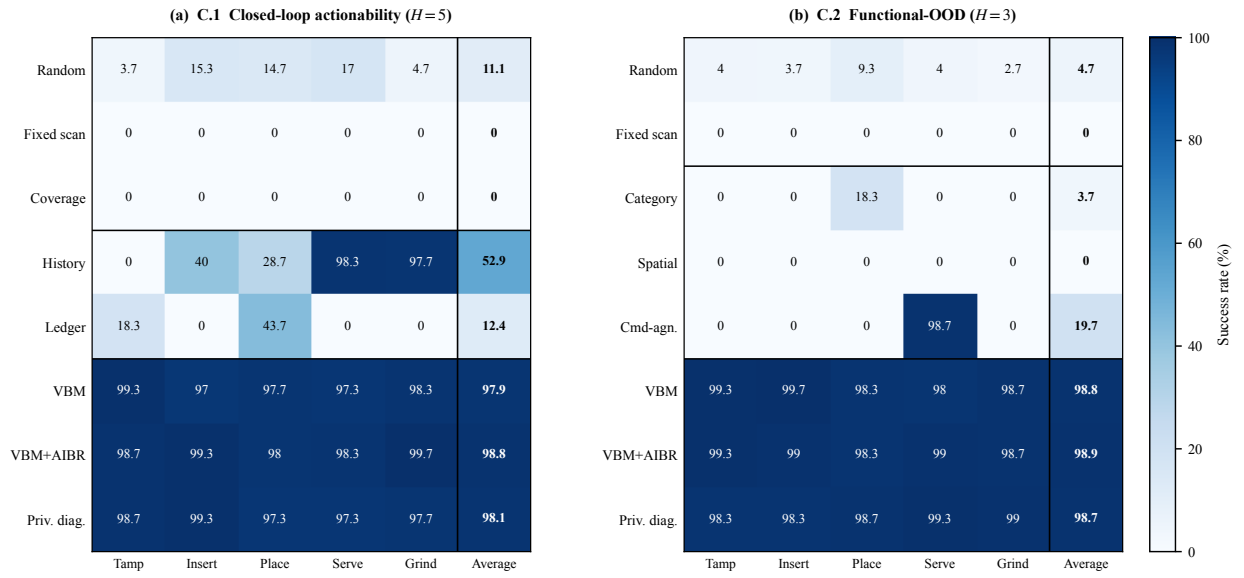



Figure 15. **Closed-loop task success by task at $H = 5$ and functional-OOD task success by task at $H = 3$.** (a,b) Task-level matrices show the five task families and the corresponding aggregate. Same-object command shifts expose category shortcuts.

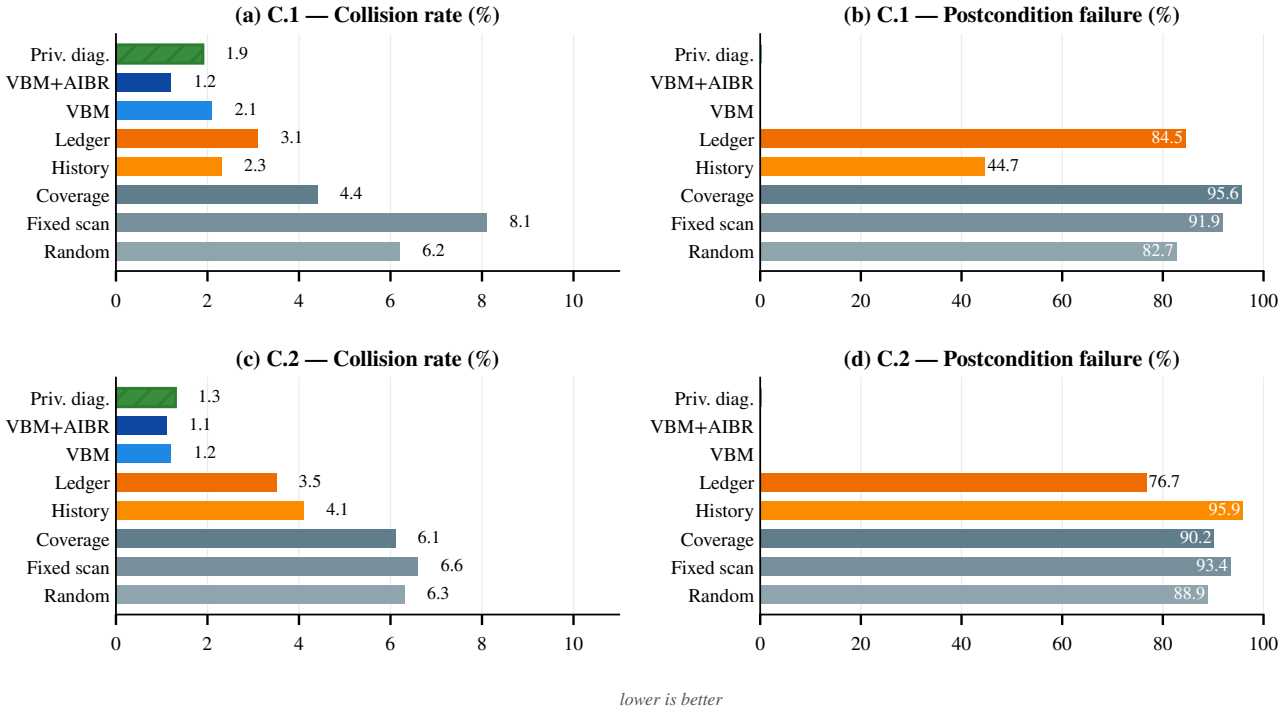



Figure 16. **Closed-loop failure metrics at $H = 5$.** Collision and postcondition failure are displayed separately. **Functional-OOD failure metrics at $H = 3$.** Failure rates expose policies that observe plausible but command-irrelevant evidence. All rates use 1,500 episodes per method/protocol as denominator; the two failure metrics are nonexclusive.

**Protocol-separation rules.** The following records are never silently merged: (i) the 180-case/domain Drawer/Microwave check and the separate 360-case/domain learned-selector branch; (ii) the C.1 ablation-run full-method value 97.60% and the 98.8% primary closed-loop value; (iii) the fixed privileged diagnostic and an oracle upper bound; (iv) QA accuracy and manipulation postcondition success; and (v) old task-family bootstrap intervals and the final paired 500-case-cluster bootstrap intervals.

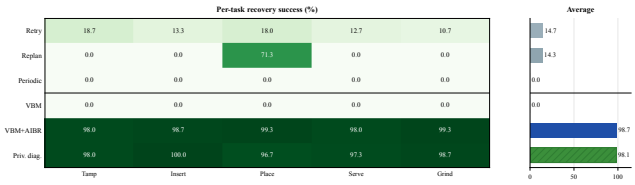



Figure 17. **Recovery after mismatch by task.** AIBR is evaluated as a repair diagnostic rather than the main source of primary-split gains. Denominators are 150 episodes per task/method and 750 per displayed method average.

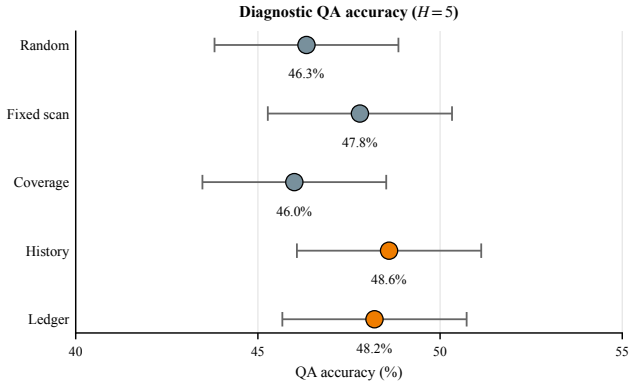



Figure 18. **VLM diagnostic QA accuracy at $H = 5$.** QA behavior is reported as a diagnostic of whether selected images contain answerable spatial-functional information; it is not treated as the main manipulation result. Each point reports `correct_count`/$n$ with $n = 1{,}500$; the normal-approximation 95% binomial half-width is about 2.5 points. No random-answer chance level is defined; Random denotes random view selection.

| Benchmark component | Fixed and auditable element |
|---|---|
| Task-conditioned actionable belief | Evidence is judged by whether it makes the current command executable, not by generic scene QA alone. |
| Frozen RGB candidate views | Every method receives the same case-level candidate set; unselected endpoint renders and hidden labels are unavailable before action. |
| Hidden postcondition evaluator | Success, collision, and postcondition failure come from hidden simulator state after execution. |
| Shared scripted skills | All methods call the same low-level primitives; differences arise from evidence selection and belief revision. |
| Five-task case structure | Five tasks, 50 scene groups per task, two variants per group, and repeated inference/evaluation records. |
| C.1/C.2 protocols | C.1 tests closed-loop actionability; C.2 tests same-object functional-OOD command shifts. |
| Strict-clean VLM extension | Seven rows enter only after the 1,500-case endpoint/JSON/exact-$H = 3$/membership/no-fallback gate. |
| Reproducibility package | Manifests, retry policy, response indexes, parsed selections, episode results, gates, audits, plotted values, and SHA256 records are packaged or referenced. |

Table 18. **Benchmark contribution checklist.** Reviewer-facing index of fixed and auditable components.

**Source-record inventory.** The reconciled numerical sources are `data/vlm_strict1500_overall.csv`, `data/vlm_strict1500_by_task.csv`, `data/c1_budget.csv`, `data/c2_budget.csv`, `data/c1_ablation.csv`, `data/c2_ablation.csv`, `data/c1_ablation_tasks.csv`, and `data/c2_ablation_tasks.csv`. The editable main and supplement sources are packaged with these relative records; absolute machine paths are not required to reproduce the paper tables.